

Genetic report for longevity

Coagulation, Thrombosis and Veins

Thrombophilias, coagulation factors and natural anticoagulants, ABO group and VIII factor, fibrinolysis, varices, phlebitis and hemorrhoids, anticoagulants

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The risk of venous thrombosis increases with age (about 1 in 1000 per year at 40 years, 1 in 100 at 80) and the wall of veins loses elasticity and valves. The Advanced Cardiovascular Report has already seen the essentials: Factor V Leiden and prothrombin G20210A negative, probable group O and a high venous thrombosis score. This report deepens: two more thrombosis scores, the VIII factor score, the natural anticoagulant loci (protein C, antithrombin), fibrinogen and fibrinolysis, veins (varicose veins, phlebitis, hemorrhoids) and pharmacogenetics of anticoagulants. Platelets are in the report Red Globules, Hemoglobin and Platelets.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

30

variants

21

in the chip

5

imputed
(DR2 $\geq$ 0.8)

4

not found

11

Scores with
percentile

Overall reading. No major thrombophilia, but with a moderate polygenic predisposition to venous thrombosis, partially compensated by the blood group. Against: the venous thrombosis consensus is at the 76th percentile (single scores at the 86th and 84th; the third, of deep venous thrombosis, in the 54); F11 TT (higher XI factor), 1 allele FGG, FGA and PROCr 219Gly, ZFPM2 AA; and the fibrinogen score is at the 92nd percentile. In favor: Factor V Leiden and prothrombin G20210A negative; probable group O, with the VIII factor score at the 15th percentile; KNG1, STXBP5 and VWF without risk alleles. In veins: varices in 64 (typical), phlebitis in 21 and hemorrhoids in 29, with PIEZO1 and CASZ1 in favor. In numbers: with a basal risk of about 1 in 1000 per year at this age, this profile corresponds to about 1.5–2 times this value. The risk is concentrated in concrete situations (long flights, surgery, immobilization, dehydration), and it is there that it is worth acting. If you ever need anticoagulation, pharmacogenetics counts: CYP2C9 *1/*3 and VKORC1 TT give much sensitivity to coumarins (prefer a direct oral anticoagulant); CYP2C19 *1/*1 leaves clopidogrel with normal efficacy.

Favourable

- **VIII factor (10 loci):** the 15th percentile
- **Phlebitis and thrombophlebitis:** the 21st percentile
- **F5 (Leiden) rs6025 CC:** Factor V Leiden (R506Q): resistance to activated protein C; 3–7 times more thrombosis per copy.
- **F2 (G20210A) rs1799963 GG:** Prothrombin G20210A: more prothrombin; 2–3 times more thrombosis.
- **KNG1 rs710446 TT:** Kininogen HMW (contact route, Ile581Thr).
- **ABO (non-O) rs505922 TT:** Tag of the non-O group: plus factor VIII and von Willebrand
- **STXBP5 rs1039084 GG:** STXBP5: von Willebrand secretion by endothelium.
- **VWF rs1063856 TT:** VWF Thr789Ala: von Willebrand levels.
- **PIEZO1 rs2911463 AA:** PIEZO1: mechanical sensor of venous and lymphatic valves.
- **CASZ1 rs11121615 TT:** CASZ1: locus of varices.
- **TSPAN15 rs17490626 CG:** TSPAN15: locus of venous thrombosis.

Unfavourable

- **Venous thrombosis (297 variants):** the 86th percentile (consensus p76)
- **Venous thrombosis (273 loci, recent score):** the 84th percentile
- **Fibrinogen (levels):** the 92nd percentile
- **F11 rs2289252 TT:** Higher XI factor; 1.2–1.4 times more thrombosis by allele
- **FGG rs2066865 GA:** Gamma fibrinogen: A allele low to gamma-line form (antithrombotic).
- **PROCcr rs867186 AG:** EPCR Ser219Gly: more soluble EPCR and less activated C protein; more venous thrombosis (and more VII factor).
- **CYP2C9 rs1057910 AC:** CYP2C9 *3: less metabolism of warfarin and acenocoumarol.
- **VKORC1 rs9923231 TT:** VKORC1 -1639G>A (T in strand read): more sensitivity to coumarins.
- **ZFPM2 rs6993770 AA:** ZFPM2: locus of venous thrombosis.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 21 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Polygenic score update (03-10-2026). The scores were recalculated with more recently published scores (same method and same reference population). The percentiles cited in this report already use the consensus (median of the scores); where a number refers to a single score, this is stated. What changed:

- Venous thrombosis: consensus **p76** (previously p86). Moves from high to above average; the practical rules stay the same.

Thrombophilia and coagulation factors

Major thrombophilias (Factor V Leiden, prothrombin G20210A) multiply the risk 2–7 times. The common variants of factor XI, fibrinogen, protein C (PROCR), antithrombin (SERPINC1) and kininogen (KNG1) add risks of 1.1–1.4 each.

Ageing mechanisms: Vascular Aging; Inter cellular Communication

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Venous thrombosis (297 variants)	Unfavourable	the 86th percentile (consensus p76) · z=1.17	Higher score : higher risk. Already seen in Advanced Cardiovascular. PGS000043 (PGS Catalog; Pruning + Clumping; Venous thromboembolism; PGP000030, PMID 31676865). Variants with genotype: 265 of 297 (covering of 84.0% coefficients).
Venous thrombosis (273 loci, recent score)	Unfavourable	the 84th percentile · z=1.04	Higher score : higher risk. PGS003456 (PGS Catalog; Genome-wide signifiers; Venous thromboembolism; PGP000449, PMID 36716319). Variants with genotype: 255 of 273 (covering of 88.0% coefficients).
Deep vein thrombosis (1773 variants)	Typical / intermediate	the 54th percentile · z=0.03	Higher score : higher risk. PGS001266 (PGS Catalog; penalized regression; Deep vein thrombosis (diagnosed by doctor); PGP000244, PMID 35324888). Variants with genotype: 1522 of 1773 (covering of 86.1% coefficients).
Fibrinogen (levels)	Unfavourable	the 92nd percentile · z=1.55	Higher score = more fibrinogen. GWAS Catalog loci score ($p \leq 10^{-6}$; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 34 of 35 loci with genotype. Studies: Danik JS (Fibrinogen); Pietzner M (Fibrinogen beta chain levels); Sabater-Lleal M (Fibrinogen); from Vries PS (Fibrinogen levels). Variants with genotype: 34 of 35 (covering of 97.1% coefficients).
D-dimers	Typical / intermediate	the 70th percentile · z=0.54	Higher score = D-dimers. Loci score of GWAS Catalog ($p \leq 510^{-8}$; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 5 of 5 loci with genotype. Studies: Pietzner M (D-dimer levels); Sherman BT (D-dimer levels in HIV infection); Smith NL (D-dimer levels). Variants with genotype: 5 of 5 (covering of 100.0% coefficients).
Pulmonary embolism	Typical / intermediate	the 64th percentile · z=0.37	Higher score : higher risk. Loci score of GWAS Catalog ($p \leq 510^{-8}$; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 28 of 34 loci with genotype. Studies: Munsch G (Venous thromboembolism recurrence in pulmonary embolism as first VTE); Verma A (Blood clot / pulmonary embolism / deep vein thrombosis (DVT)); Verma A (Pulmonary embolism and infarction, acute (PheCode 415.11)); Zhang Z (Pulmonary embolism). Variants with genotype: 28 of 34 (covering of 82.4% coefficients).

Favourable

- F5 (Leiden) rs6025 CC
- F2 (G20210A) rs1799963 GG
- KNG1 rs710446 TT
- TSPAN15 rs17490626 CG

Unfavourable

- Venous thrombosis (297 variants) (the 86th percentile; consensus p76)
- Venous thrombosis (273 loci, recent score) (the 84th percentile)
- Fibrinogen (levels) (the 92nd percentile)
- F11 rs2289252 TT
- FGG rs2066865 GA
- FGA rs6050 TC
- PROCR rs867186 AG
- ZFPM2 rs6993770 AA

Favourable variants (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
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F5 (Leiden)	rs6025 1:169549811 · C/A/T	CC	chip	T (0.0; 1%)	strong	Factor V Leiden (R506Q): resistance to activated protein C; 3–7 times more thrombosis per copy. Negative for Factor V Leiden.
F2 (G20210A)	rs1799963 11:46739505 · G/A	GG	chip	A (0.0; 1%)	strong	Prothrombin G20210A: more prothrombin; 2–3 times more thrombosis. Negative for prothrombin G20210A.
KNG1	rs710446 3:186742138 · T/C/G	TT	chip	C (0.0; 42%)	moderate	Kininogen HMW (contact route, Ile581Thr).
TSPAN15	rs17490626 10:69458890 · G/A/C	CG	chip	G (1.0; 87%)	moderate	TSPAN15: locus of venous thrombosis. Meaning: GWAS Catalog: Blood clot / pulmonary embolism / deep vein thrombosis (DVT) (Verma A 2024; p=8e-28); G ↑ allele

Unfavourable variants (5)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
F11	rs2289252 4:186286227 · C/T	TT	chip	T (2.0; 39%)	strong	Higher XI factor; 1.2–1.4 times more thrombosis by allele. The XI factor is the target of new anticoagulants (abelacimab, asundexian).
FGG	rs2066865 4:154604124 · G/A/C	GA	chip	A (1.0; 22%)	moderate	Gamma fibrinogen: A allele low to gamma-line form (antithrombotic).
FGA	rs6050 4:154586438 · T/A/C	TC	chip	C (1.0; 24%)	weak	Fibrinogen alpha Thr312Ala: more rigid clot.
PROCR	rs867186 20:35176751 · A/G	AG	chip	G (1.0; 9%)	moderate	EPCR Ser219Gly: more soluble EPCR and less activated C protein; more venous thrombosis (and more VII factor).
ZFPM2	rs6993770 8:105569300 · A/T	AA	imputation DR2=0.99	A (2.0; 70%)	moderate	ZFPM2: locus of venous thrombosis. Meaning: GWAS Catalog: Venous thromboembolism (Thibord F 2022; p=4e-48); Allele A ↑

Typical / intermediate variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
F11	rs2036914 4:186271327 · T/A/C/G	TC	chip	C (1.0; 53%)	moderate	XI factor: second independent signal.
SERPINC1	rs2227589 1:173917078 · C/A/T	CC	chip	T (0.0; 10%)	moderate	Antithrombin: a common variant that slightly lowers its levels.
SLC44A2	rs2288904 19:10631494 · A/G/T	GG	chip	G (2.0; 76%)	moderate	SLC44A2: interaction of neutrophils with von Willebrand. Meaning: GWAS Catalog: Venous thromboembolism (Klarin D 2019; p=4e-24); G ↑ allele

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
F13A1	rs5985 6:6318562 · C/A/T	AC	chip	—	weak	XIII Val34Leu factor: Leu activates faster; small and inconsistent associations. Information.
THBD		GG	imputation DR2=0.96	—	weak	Thrombomodulin A455V.

rs1042579
20:23048087 ·
G/A/T

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
GP6	rs1654425 — · —	—	SNP not found in the data	—	moderate	GP6: platelet receptor of collagen. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Group ABO, factor VIII and von Willebrand

The non-O group has about 25% plus factor VIII and von Willebrand and about 1.5–2 times more thrombosis. Factor VIII and von Willebrand rise with age and inflammation; STXBP5, VWF and CLEC4M touch their levels.

Ageing mechanisms: Vascular Aging; Inflammaging

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
VIII factor (10 loci)	Favourable	the 15th percentile · $z=-1.12$	Higher score = more VIII factor. PGS003330 (PGS Catalog; Genome-wide signifier SNPs; Factor VIII levels; PGP000395, PMID 36313186). Variants with genotype: 9 of 10 (covering of 95.4% coefficients).
Platelet count (739 loci)	No judgement (informative)	the 72nd percentile · $z=0.55$	Information; detail in the platelet report. PGS000186 (PGS Catalog; Genome-wide signifiers; Platelet count; PGP000078, PMID 32888494). Variants with genotype: 578 of 739 (covering of 80.9% coefficients).

Favourable

- VIII factor (10 loci) (the 15th percentile)
- ABO (non-O) rs505922 TT
- STXBP5 rs1039084 GG
- VWF rs1063856 TT

Unfavourable

None.

Favourable variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ABO (non-O)	rs505922 9:133273813 · C/T	TT	chip	C (0.0; 37%)	moderate	Tag of the non-O group: plus factor VIII and von Willebrand. TT = probable group O.
STXBP5	rs1039084 6:147314277 · A/G/T	GG	imputation DR2=1.00	A (0.0; 48%)	moderate	STXBP5: von Willebrand secretion by endothelium.
VWF	rs1063856 12:6044368 · T/C/G	TT	chip	C (0.0; 37%)	moderate	VWF Thr789Ala: von Willebrand levels.

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
ABO	rs8176719 9:133257522 · -/C	TT	chip	—	moderate	Deletion of the O allele (indel; rarely read by chip). The chip returns TT in a deletion: non-interpretable code. Group O is inferred by rs505922.
SERPINE1	rs2227631 7:101126257 · A/G	AA	chip	—	weak	PAI-1 (fibrinolysis): tag close to 4G/5G, which is an indel and is not read. Information.

Veins: varices, phlebitis and hemorrhoids

Varices depend on the venous wall and valves (PIEZO1, CASZ1, EBF1). The superficial phlebitis shares loci with thrombosis; hemorrhoids share loci with the matrix and intestinal motility.

Ageing mechanisms: Remodelling of the extracellular matrix; Fragmentation of Elastin

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Varieties (15 748 variants)	Typical / intermediate	the 64th percentile · z=0.33	Higher score : higher risk. Already seen in the Extracellular Matrix report. PGS001845 (PGS Catalog; Penalized regression; Varicose vein; PGP000263, PMID 34995502). Variants with genotype: 15341 of 15748 (covering of 96.8% coefficients).
Phlebitis and thrombophlebitis	Favourable	the 21st percentile · z=-0.80	Higher score : higher risk. PGS001844 (PGS Catalog; Penalized regression; Phlebitis and thrombophlebitis; PGP000263, PMID 34995502). Variants with genotype: 425 of 431 (covering of 99.6% coefficients).
Hemorrhoids (5434 variants)	Typical / intermediate	the 29th percentile · z=-0.57	Higher score : higher risk. PGS001846 (PGS Catalog; Penalized regression; Hemorrhoids; PGP000263, PMID 34995502). Variants with genotype: 5317 of 5434 (covering of 97.2% coefficients).

Favourable

- Phlebitis and thrombophlebitis (the 21st percentile)
- PIEZO1 rs2911463 AA
- CASZ1 rs11121615 TT

Unfavourable

None.

Favourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
PIEZO1	rs2911463 · 16:88769137 · G/A/C/T	AA	imputation · DR2=0.97	G (0.0; 30%)	moderate	PIEZO1: mechanical sensor of venous and lymphatic valves. Meaning: GWAS Catalog: Veins varicose (PheCode 454) (Verma A 2024; p=7e-12); G ↑ allele
CASZ1	rs11121615 · 1:10765520 · C/T	TT	chip	C (0.0; 34%)	moderate	CASZ1: locus of varices. Meaning: GWAS Catalog: Varicose vein (Sakaue S 2021; p=1e-103); allele T ↓

SNP not found in the data (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
EBF1	rs11135046 · — · —	—	SNP not found in the data	—	moderate	EBF1: locus of varices. No major variant of GWAS Catalog for this gene with usable genotype and sense.
SOX17	— · — · —	—	SNP not found in the data	—	moderate	SOX17: venous endothelium and valves. No major variant of GWAS Catalog for this gene with usable genotype and sense.
HFE	— · — · —	—	SNP not found in the data	—	moderate	HFE: locus of varices in a large reference cohort. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Anticoagulants and antiplatelet agents

If one day needs anticoagulation: CYP2C9, VKORC1 and CYP4F2 (coumarins), CES1 (dabigatran), CYP2C19 (clopidogrel). These genotypes are already in the pharmacogenetic profile.

Ageing mechanisms: Nutrient Sensing Deregulation

Favourable

None.

Unfavourable

- CYP2C9 rs1057910 AC
- VKORC1 rs9923231 TT

Unfavourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CYP2C9	rs1057910 10:94981296 · A/C/G	AC	chip	C (1.0; 7%)	strong	CYP2C9 *3: less metabolism of warfarin and acenocoumarol.
VKORC1	rs9923231 16:31096368 · C/A/G/T	TT	imputation DR2=0.99	T (2.0; 39%)	strong	VKORC1 -1639G>A (T in strand read): more sensitivity to coumarins.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CYP2C19	rs4244285 10:94781859 · G/A/C/T	GG	chip	A (0.0; 14%)	strong	CYP2C19 *2: loss of function; clopidogrel less effective.

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CYP4F2	rs2108622 19:15879621 · C/A/G/T	CC	chip	—	moderate	CYP4F2 V433M: the T allele asks for more coumarin dose. Information.
CES1	rs2244613 16:55810697 · G/A/C/T	TT	chip	—	moderate	CES1: the minority allele lowers the levels of dabigatran and the risk of bleeding. Information.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Intercellular Communication	Venous thrombosis (297 variants) (the 86th percentile; consensus p76); Venous thrombosis (273 loci, recent score) (the 84th percentile); Fibrinogen (levels) (the 92nd percentile); F11 TT; FGG GA; FGA TC; PROCRA AG; ZFPM2 AA	F5 (Leiden) CC; F2 (G20210A) GG; KNG1 TT; TSPAN15 CG	Deep vein thrombosis (1773 variants) (the 54th percentile); D-dimers (the 70th percentile); Pulmonary embolism (the 64th percentile); F11 TC; SERPINC1 CC; SLC44A2 GG
Vascular Aging	Venous thrombosis (297 variants) (the 86th percentile; consensus p76); Venous thrombosis (273 loci, recent score) (the 84th percentile); Fibrinogen (levels) (the 92nd percentile); F11 TT; FGG GA; FGA TC; PROCRA AG; ZFPM2 AA	F5 (Leiden) CC; F2 (G20210A) GG; KNG1 TT; TSPAN15 CG; VIII factor (10 loci) (the 15th percentile); ABO (non-O) TT; STXBP5 GG; VWF TT	Deep vein thrombosis (1773 variants) (the 54th percentile); D-dimers (the 70th percentile); Pulmonary embolism (the 64th percentile); F11 TC; SERPINC1 CC; SLC44A2 GG
Nutrient Sensing Deregulation	CYP2C9 AC; VKORC1 TT	—	CYP2C19 GG
Fragmentation of Elastin	—	Phlebitis and thrombophlebitis (the 21st percentile); PIEZO1 AA; CASZ1 TT	Varieties (15 748 variants) (the 64th percentile); Hemorrhoids (5434 variants) (the 29th percentile)
Inflammaging	—	VIII factor (10 loci) (the 15th percentile); ABO (non-O) TT; STXBP5 GG; VWF TT	—
Remodelling of the extracellular matrix	—	Phlebitis and thrombophlebitis (the 21st percentile); PIEZO1 AA; CASZ1 TT	Varieties (15 748 variants) (the 64th percentile); Hemorrhoids (5434 variants) (the 29th percentile)

Practical part

Ideas for coagulation and veins, crossed with the genetic plan v2 and with the current medication (dapagliflozin, retatrutide, isotretinoin, rapamycin, tadalafil, telmisartan) and with the fish oil that already takes. The current stack remains as it is; everything is suggestion (not applied). There is no indication for anticoagulation or antiaggregation because of these genes: the objective is to reduce the risk in situations of higher risk.

1. Long trips

Genetics: thrombosis p76 (consensus; single scores p86 and p84); F11 TT

In trips of more than 4 h: standing and walking every 1–2 h, twins exercises sitting, drinking water, without alcohol (no longer drink). In flights of more than 6–8 h, compression stockings below the knee (15–30 mmHg). No aspirin or heparin only for traveling.

V2 genetic plan and medication: Compatible with the plan (already indicated in Advanced Cardiovascular). • suggestion (not applied)

2. Surgery, fracture or immobilization

Genetics: Polygenic predisposition; without Leiden or G20210A

Always tell the doctor that there is moderate polygenic predisposition (F11 TT, consensus at the 76th percentile), so that prophylaxis follows the clinical risk scores with this data.

V2 genetic plan and medication: No conflict. • suggestion (not applied)

3. Daily hydration

Genetics: High genetic fibrinogen; thrombosis scores

Dapagliflozin slightly concentrates blood (sobe the hematocrit) and retatrutide reduces fluid intake. Water throughout the day, more on hot days, long training or travel.

V2 genetic plan and medication: Compatible with the plan. • suggestion (not applied)

4. Measure fibrinogen once and factor VIII

Genetics: Fibrinogen p92 score; factor VIII p15

In the next routine analysis, fibrinogen and factor VIII (and D-dimers if the doctor finds it useful) to see if the values follow the genetics. Fibrinogen rises with inflammation; factor VIII rises with age.

V2 genetic plan and medication: Adds 2 analysis to the panel. • suggestion (not applied)

5. If you need anticoagulation

Genetics: CYP2C9 *1/*3; VKORC1 TT; CES1 TT; CYP2C19 *1/*1

Prefer apixaban or rivaroxaban. With warfarin or acenocoumarol, much lower initial dose and tight INR.

Vitamin K2 does not touch the direct anticoagulants, but with a coumarin must be constant or stopped; tell the doctor who takes K2 and fish oil.

V2 genetic plan and medication: The same pharmacogenetic profile. • suggestion (not applied)

6. Veins: muscle pump and fiber

Genetics: p64 varices; p29 hemorrhoids

Walking and training the legs keeps the twin pump; the dietary fiber (the more, the better) and water protect from hemorrhoids.

V2 genetic plan and medication: Compatible with the plan. • suggestion (not applied)

Avoiding or with caution

- Testosterone, EPO or prohormones without follow-up: elevated hematocrit and risk of thrombosis. suggestion (not applied)
- Add nattokinase, lumbroquinase, ginkgo or vitamin E in high dose to fish oil: add antiplatelet effect without proven benefit. suggestion (not applied)
- Coumarines in usual dose (CYP2C9 *1/*3 + VKORC1 TT). suggestion (not applied)
- Dehydration and prolonged immobility (viages, disease), especially with dapagliflozin. suggestion (not applied)

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. The ABO O allele and the PAI-1 4G/5G are indels and are not read; the group O is inferred by a tag.
- Hereditary deficits of protein C, protein S and antithrombin are rare and private mutations: they do not see themselves as chips; they ask for blood dosage if there is thrombosis without cause.
- The thrombosis scores are relative: with a basal risk of about 1 in 1000 per year, an the 85th percentile corresponds to about 1.5–2 times this value.
- Two loci scores were calculated and not interpreted: von Willebrand (13 loci of studies with different scales, result in conflict with group O and the VIII factor score) and PAI-1 (40 coverage).
- Rapamycin was associated with venous thrombosis in transplanted patients (very different doses and contexts); isotretinoin has weak signs of pharmacovigilance. None of them change the suggestions.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local statistical imputation (with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS000043 (PGS Catalog; Pruning + Clumping; Venous thromboembolism; PGP000030, PMID 31676865 · PGS003456 (PGS Catalog; Genome-wide signifiers; Venous thromboembolism; PGP000449, PMID 36716319 · PGS001266 (PGS Catalog; penalized regression; Deep vein thrombosis (diagnosed by doctor); PGP000244, PMID 35324888 · PGS003330 (PGS Catalog; Genome-wide signifier SNPs; Factor VIII levels; PGP000395, PMID 36313186 · PGS001845 (PGS Catalog; Penalized regression; Varicose vein; PGP000263, PMID 34995502 · PGS001844 (PGS Catalog; Penalized regression; Phlebitis and thrombophlebitis; PGP000263, PMID 34995502 · PGS001846 (PGS Catalog; Penalized regression; Hemorrhoids; PGP000263, PMID 34995502 · PGS000186 (PGS Catalog; Genome-wide signifiers; Platelet count; PGP000078, PMID 32888494).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

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